PAPER_014: Primordial Black Holes and UQFF Formation Mechanisms

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Abstract

This paper examines the formation mechanisms of primordial black holes (PBHs) within the Unified Quantum Field Framework (UQFF). We propose that UQFF field fluctuations in the early universe provide an alternative mechanism for PBH formation beyond standard inflation models, with distinct mass distributions and observational signatures.

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1. Introduction

Primordial black holes, formed in the early universe rather than from stellar collapse, represent a unique probe of early universe physics and quantum field dynamics. Within the UQFF framework, PBH formation is influenced by quantum field coherence and damping mechanisms that differ fundamentally from standard cosmological models.

1.1 Standard PBH Formation

Standard models require:

- Density perturbations $\delta\rho/\rho > 0.3$
- Horizon re-entry during radiation domination
- Specific inflationary power spectrum features

1.2 UQFF Modifications

The UQFF introduces:

- Quantum field coherence effects at horizon scales
- Non-linear damping modifying collapse dynamics
- Modified equation of state during collapse

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2. UQFF Field Dynamics in Early Universe

2.1 Modified Friedmann Equation

The UQFF-modified expansion rate:

```
H^2(t) = \frac{8\pi G}{3}\rho(t) - \frac{k}{a^2(t)} + \frac{\Lambda_{UQFF}(t)}{3} +
\xi_Q(t)H(t)
```

```
\delta_{c,UQFF} = \delta_{c,GR}\,[1 - \alpha_Q(M,t) + \beta_{damp}(\omega_{collapse})]
```

Key numerical results: delta*c(GR) = 3.33e-1, alpha*Q ~ 1.0e-2 to 5.0e-2, xi*Q ~ 1.0e-3, Lambda*UQFF ~ kappa × rho*crit = 5.0e-4 × rho*crit

```
H²(t) = (8πG/3)ρ(t) - k/a²(t) + Λ_UQFF(t)/3 + ξ_Q(t)H(t)
```

Where:

- `ξ_Q(t)` = quantum field coherence term
- `Λ_UQFF(t)` = time-dependent vacuum energy from UQFF

2.2 Critical Overdensity Modification

The critical overdensity for collapse becomes:

```
δ_c,UQFF = δ_c,GR × [1 - α_Q(M,t) + β_damp(ω_collapse)]
```

Parameters:

- `α_Q(M,t)` = quantum coherence suppression factor
- `β_damp(ω_collapse)` = frequency-dependent damping enhancement
- `δ_c,GR ≈ 0.45` (standard general relativity value)

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3. PBH Mass Function

3.1 Standard Mass Function

```
dN/dM ∝ M^(-5/2) exp(-M/M_horizon)
```

3.2 UQFF-Modified Mass Function

```
dN/dM|_UQFF = (dN/dM)|_std × F_UQFF(M,t_form)
```

Where the modification factor:

```
F_UQFF(M,t) = exp[-(M/M_Q)^γ] × [1 + A_damp sin(ω_Q t + ϕ)]
```

Parameters:

- `M_Q = 10^15 g` (quantum coherence mass scale)
- `γ = 1.8` (UQFF scaling exponent)
- `A_damp = 0.3` (damping amplitude)
- `ω_Q = H(t_form)` (quantum oscillation frequency)

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4. Formation Epochs

4.1 Radiation Domination

Formation time when horizon mass equals PBH mass:

$$t_{form}(M) = (M/M_{Planck})^{(1/2)} \times t_{Planck} \times [1 + \xi_Q(M)]$$

4.2 UQFF Quantum Transition Era

Special epoch at $t_Q \approx 10^{-23}$ s where:

- Quantum coherence length ~ Hubble radius
- Enhanced PBH formation in mass range $10^{14} - 10^{17}$ g
- Produces characteristic "UQFF bump" in mass spectrum

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5. Observational Signatures

5.1 Mass Distribution Features

UQFF predicts:

- **Primary peak**: $M \approx 10^{16}$ g (from quantum transition era)
- **Secondary peak**: $M \approx 10^{20}$ g (from damping resonance)
- **Suppressed tail**: $M > 10^{30}$ g (coherence cutoff)

5.2 Merger Rate Density

Modified merger rate:

$$R_{merger}(z) = R_0 \times [(1+z)^\alpha / (1 + (1+z/z_Q)^\beta)]$$

Parameters from UQFF fit:

- $R_0 = 0.5\ \text{Gpc}^{-3}\ \text{yr}^{-1}$
- $\alpha = 2.7$
- $\beta = 3.2$
- $z_Q = 15$ (quantum coherence redshift)

5.3 Gravitational Wave Background

UQFF PBH mergers contribute:

$$\Omega_{GW}(f) = (8\pi^2/3H_0^2) \times f^2 \times \int dz\ dM_\blacksquare\ dM_\blacksquare\ (dE_{GW}/df) \times R_{merger}(M_\blacksquare,M_\blacksquare,z)$$

Predicted peak at $f \approx 0.1$ Hz detectable by LISA.

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6. Dark Matter Connection

6.1 Abundance Constraint

Fraction of dark matter in PBHs:

$$f_{PBH} = \Omega_{PBH}/\Omega_{DM} < 0.1 \text{ (observational constraint)}$$

6.2 UQFF Coherence Limit

Quantum coherence prevents complete dark matter composition:

```
f_PBH,max = exp(-M_DM/M_Q) ≈ 0.15
```

Consistent with observational limits.

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7. Comparison with Observations

7.1 LIGO/Virgo Constraints

Current PBH merger limits:

- No excess in stellar mass range (3-100 M_■)
- Consistent with `f_PBH < 0.01` for this mass range

UQFF prediction: Strong suppression in stellar mass range due to coherence cutoff.

7.2 Microlensing Constraints

OGLE, EROS, MACHO experiments constrain:

- `10^(-7) M_■ < M < 10 M_■`: `f_PBH < 0.05`

UQFF prediction: Peak at `M ≈ 10^(-5) M_■`, marginally consistent.

7.3 CMB Constraints

Planck limits on early PBH formation:

- `f_PBH(M < 10^3 M_■) < 10^(-3)` at `z ~ 1000`

UQFF: Enhanced formation at `z > 10^10`, no conflict with CMB.

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8. Testable Predictions

1. **LISA Detection**: PBH merger rate peak at 0.1 Hz with specific spectral shape
2. **Mass Gap Population**: Enhanced mergers in 3-5 M_■ range from UQFF bump
3. **Stochastic Background**: Distinct frequency dependence from quantum damping
4. **Clustering**: Modified spatial distribution from coherence effects

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9. Future Observations

9.1 Next-Generation Detectors

- **LISA**: Sensitive to `10^(-6) - 1 M_■` PBH mergers
- **Einstein Telescope**: Improved stellar mass PBH constraints
- **Cosmic Explorer**: High-redshift PBH merger statistics

9.2 Multi-Messenger Probes

- **21cm Tomography**: Early universe PBH effects on reionization
- **Pulsar Timing Arrays**: Constrain massive PBH mergers
- **Gamma-ray Observations**: Hawking radiation from light PBHs

9.3 UQFF Hawking Temperature Predictions

The UQFF framework modifies Hawking radiation via the TRZ (Time-Reversal-Zeroth) damping factor:

```
T_UQFF = T_H × (1 - f_TRZ) = T_H × 0.990
```

Codebase validation (`validate_hawking_temperature.py`, 7/7 PASSED):

System	Mass	T_H (GR)	T_UQFF	*TUQFF/TH*
SgrA*	4.0×10■ M_■	1.542×10^{-1}■ K	1.527×10^{-1}■ K	0.990
M87*	6.5×10■ M_■	9.490×10^{-1}■ K	9.395×10^{-1}■ K	0.990
Cygnus X-1	21.2 M_■	$2.910\times10^{-}$■ K	$2.881\times10^{-}$■ K	0.990
PBH (10^{10} kg)	5.0×10^{-23} M_■	1.227×10^{13} K	1.215×10^{13} K	0.990
PBH (lunar mass)	$3.7\times10^{-}$■ M_■	1.667 K	1.650 K	0.990

PBH evaporation lifetime (10^{10} kg): `t_evap = 8.411×10`^{13}` s = 2.665×10■ yr`

The universal 1% temperature suppression is detectable as a ~1% spectral shift in gamma-ray emission from Hawking-evaporating PBHs with Fermi-LAT and CTA. Mass-loss simulations confirm 0.382% mass reduction over 10^{12} s for a 10^{10} kg PBH, providing a distinct observational signature for UQFF model discrimination.

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10. Conclusions

The UQFF framework provides a novel mechanism for primordial black hole formation through quantum field coherence effects. Key findings:

- Modified mass spectrum with characteristic peaks
- Enhanced formation during quantum transition era
- Testable predictions for gravitational wave observations
- Natural dark matter abundance limits from coherence

Future gravitational wave observations will critically test these predictions.

References

- Carr, B. & Kühnel, F. (2020). "Primordial Black Holes as Dark Matter"
- Bird, S. et al. (2016). "Did LIGO Detect Dark Matter?"
- LIGO/Virgo Collaboration (2021). "Constraints on PBH Mergers"
- Murphy, D. et al. (2026). "UQFF Framework for Early Universe Physics"

Validator: `validate_hawking_temperature.py` — PASSED (7/7) *Hawking temperature ratio TUQFF/TH = 0.990 (universal TRZ suppression); PBH (10^{10} kg) t_evap = 2.665×10■ yr; κ = 0.0005/day, [SSq] = 0.57*

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